

Electric Vehicle Charging Systems

Converter Architectures, Control
Optimization,
and PV-Integrated Energy Management

Authors :

Mubaarak A.A. Saif

Salah F. S. Saeed

Abdullah Haidar

Alzahraa F. M. Almakhathay

Malak M. N. Al-Koshab

Nora H. A. Qasem

Presented by

Alzahraa F. M. Almakhathay



Na'mah Rassam Center
for Research, Studies, Technology & Innovation



EV Charging

Optimization

PV Integration

Introduction

Why the Three Research Layers Must Be Read Together

- 1  Modern EV chargers require fast charging, wide-voltage operation, bidirectional power flow, and grid interaction.
- 2  Research has evolved from converter hardware to advanced control and PV-integrated energy management.
- 3  Converter architecture determines control strategies, while PV and grid integration influence overall system dynamics.

Central thesis: deployment readiness requires cross-layer, not isolated, design.



Review Scope & Objective

From fragmented research domains to an integrated analytical view

Analytical Problem

The evidence is dispersed across three technically coupled domains:

- 1 Converter and charger architecture.
- 2 Control and optimization.
- 3 PV-integrated charging and energy management.

Treating these domains independently obscures cross-layer constraints and deployment trade-offs.

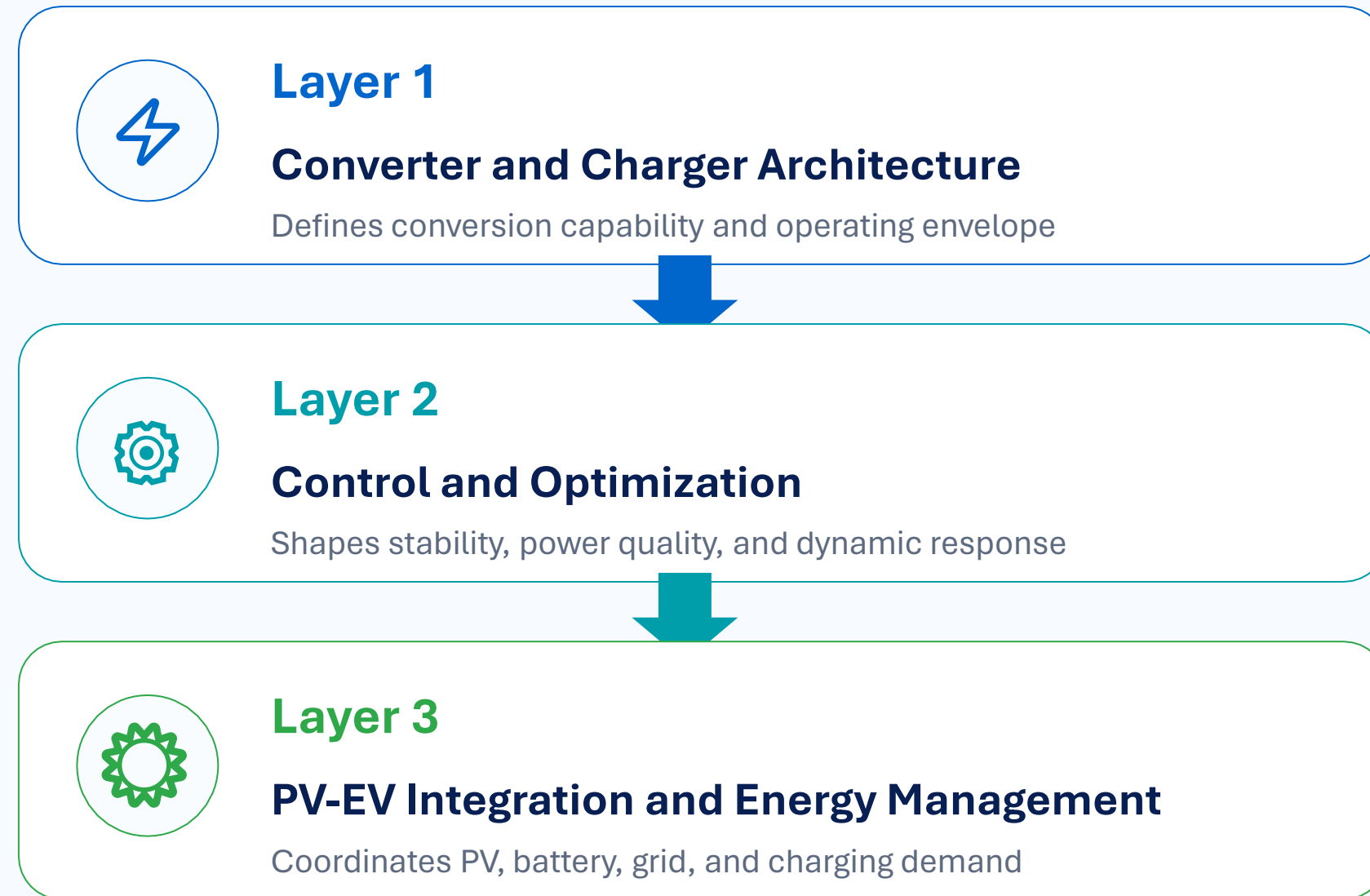
Review Objective

- 1 Synthesize the three evidence streams.
- 2 Explain their technical interdependencies.
- 3 Identify cross-group gaps and limitations.
- 4 Define evidence-based research priorities.

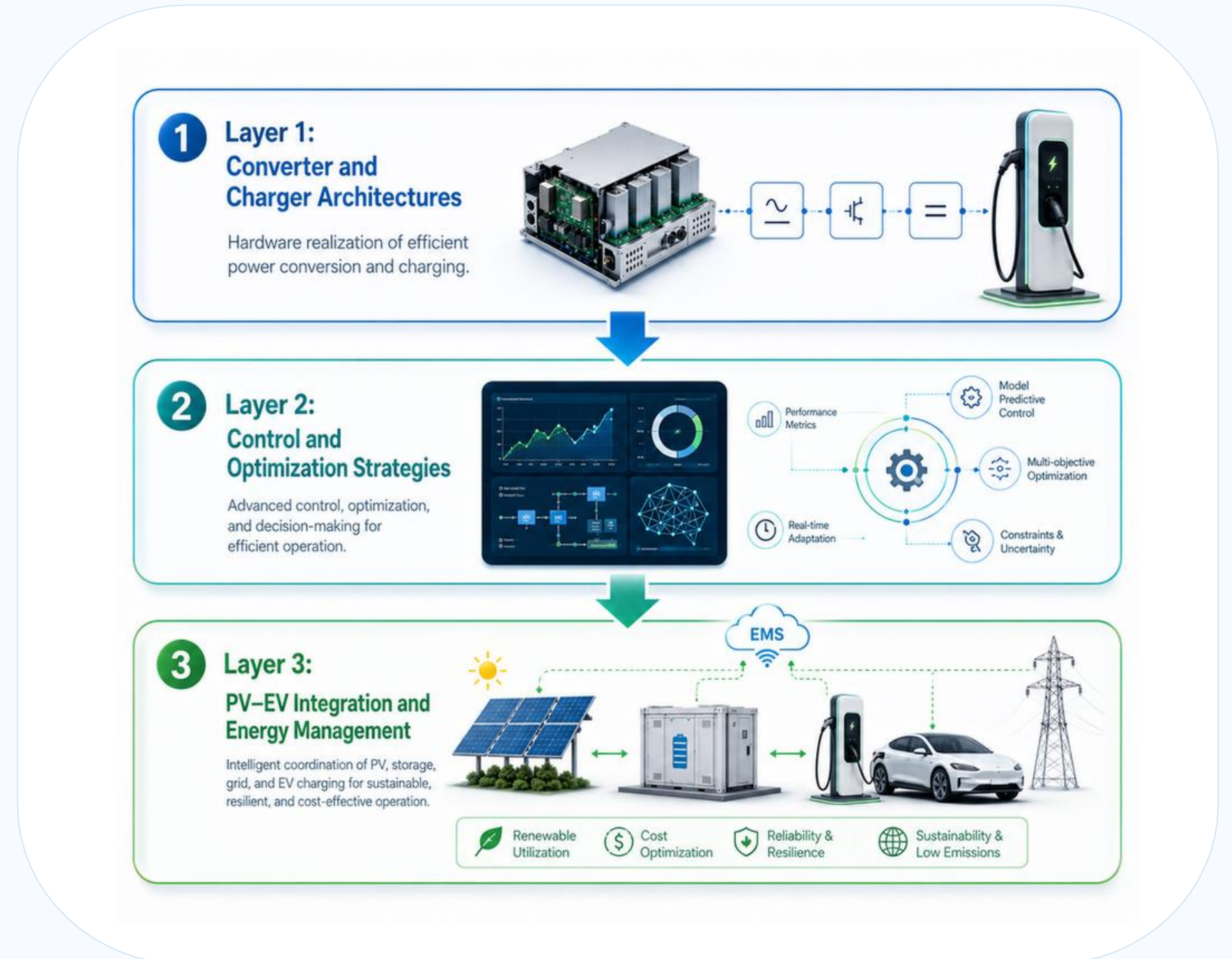


A Three-Layer Framework Connects the Field

Hardware realization → control intelligence → integrated energy coordination



The framework reveals that advances at one layer can constrain or destabilize the others.

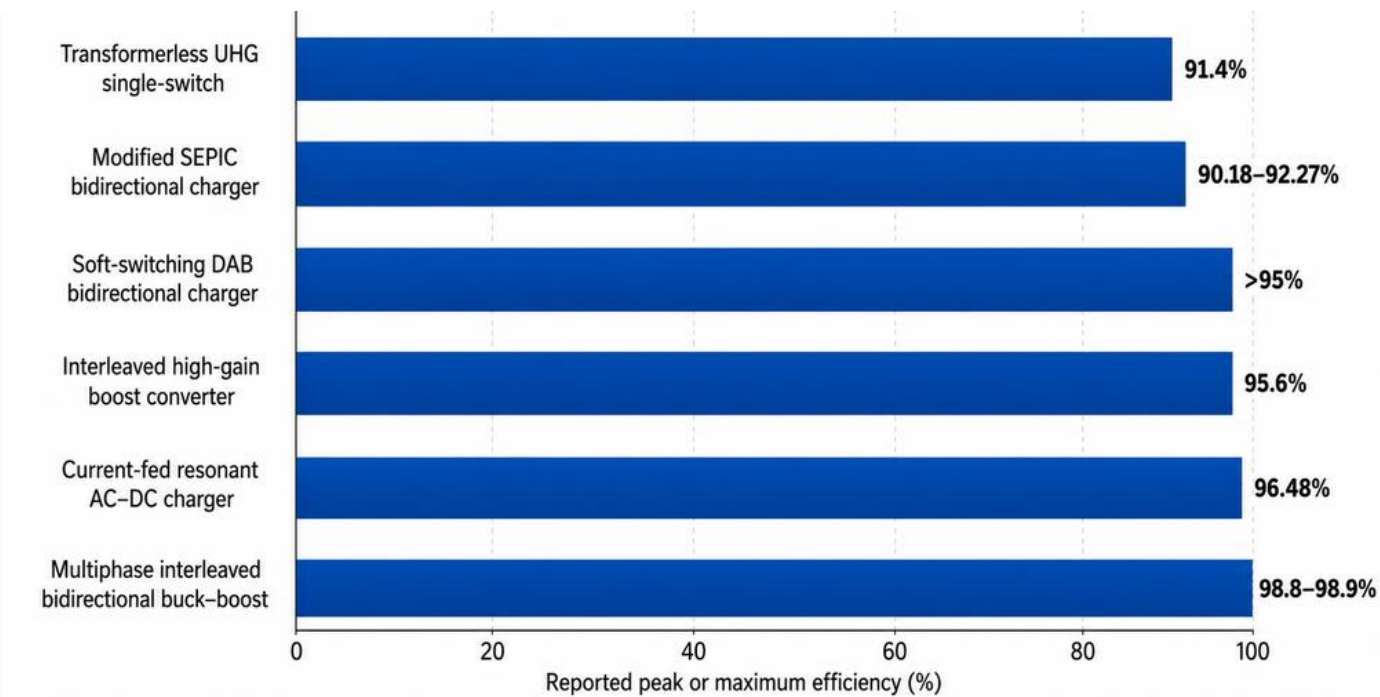


Layer I: Architecture Expands Capability

Converter and Charger Architectures

TABLE I
COMPRESSED COMPARISON OF CONVERTER AND CHARGER ARCHITECTURE STUDIES

Ref.	Year	Topology	Contribution	Validation	Performance	Limitation
[8]	2025	Single-stage CF resonant AC-DC charger	400/800 V resonant fast charger with integrated PFC + soft switching	Sim. + HIL	(η_{pk}) 96.48%; PF 0.998–0.999; THD 2.8%	No prototype; 800 V only simulated
[9]	2024	3 ϕ VSC + SS-DAB	Soft-switching bidirectional OBC for G2V/V2G	Sim.	Battery $(\eta) > 95\%$	No experiments; optimization not explicitly supported
[10]	2025	Quadratic boost + DAB + T-type inverter	Isolated two-stage bidirectional FCBEV charger	Sim.	THD $< 2.35\%$; $(V_{dc}) \approx 700$ V	No hardware; incomplete GA reporting
[11]	2025	VSC + mod. SEPIC	Wide-range bidirectional charger for low-voltage batteries	Sim. + exp.	(η) 90.18–92.27%; ripple $< 10\%$	Low-power only (1.1 kW); no formal optimization
[12]	2025	Multiphase interleaved bidir. buck-boost	Interleaving + intraleaving + phase shedding for loss minimization	Sim. + exp.	(η_{pk}) 98.8–98.9%; light-load (η) 97–99%	Higher conduction/RMS losses; not ideal at high power
[13]	2024	Interleaved high-gain non-isolated boost	Ripple/stress reduction via phase-shifted interleaving	Sim. + exp.	(η_{pk}) 95.6%; gain ≈ 4 ; low ripple	Thermal/parasitic sensitivity; limited scalability
[14]	2024	Transformerless UHG single-switch buck-boost	Ultra-high-gain reduced-stress single-switch design	Sim. + exp.	(η_{max}) 91.4%; $(V_o) \approx 720$ V	High passive count; open-loop only



Peak Efficiency Across Representative EV-Charger Architectures



Evidence Limits

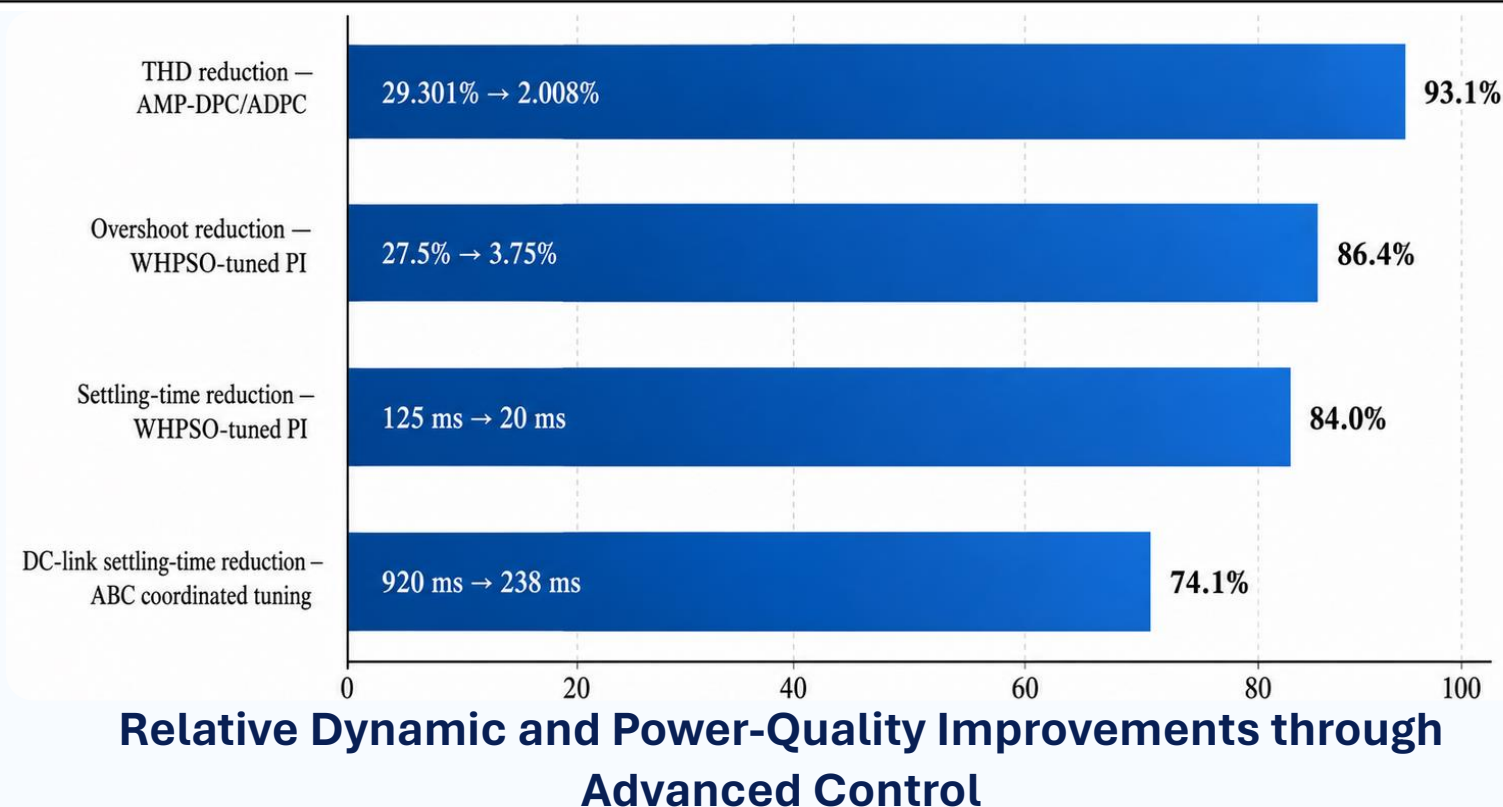
- 1 Simulation/HIL outpaces prototype evidence.
- 2 Power scaling and thermal sensitivity remain open.
- 3 Efficiency gains trade against RMS/conduction/passive burden.

Layer II: Control Drives Dynamic Gains

Control and optimization

TABLE II
COMPRESSED COMPARISON OF CONTROL AND OPTIMIZATION STUDIES FOR EV CHARGING SYSTEMS

Ref.	Year	Controller Family	Optimization / Control Strategy	Principal Control Contribution	Validation Level	Key Reported Improvement	Principal Limitation
[15]	2025	PID	GWO-optimized PID	Adaptive controller tuning for improved inverter regulation and power quality in a grid-connected PV system	Simulation only	THD reduced to 3.7%; settling time 0.035 s; PF ≈ 1	No hardware validation
[16]	2024	AMP-DPC + ADPC	Adaptive predictive/direct power control	Dual control strategy for bidirectional EV charger power-quality enhancement across G2V/V2G/V2L modes	Experimental prototype	THD reduced from 29.301% to 2.008%; settling time improved to 5–8 ms	No formal optimization framework; limited to 0.5 kW prototype
[17]	2024	PI	AIChOa-based PI tuning	Hybrid metaheuristic tuning of charger control gains for improved V2G/G2V performance	Simulation only	THD reduced to 4.56%; improved convergence and reduced error metrics	Simulation only; objective function remains simplified
[18]	2025	Robust nonlinear / sliding-mode-based coordinated control	Optimal coordination methodology	Robust coordinated control for bidirectional BEV chargers under V2G/G2V operation	Simulation + CHIL	Improved tracking and reduced DC-bus error relative to baseline controllers	Optimization settings and statistical validation not fully reported
[19]	2024	Cascaded PI	WHPSO-based tuning	Metaheuristic optimization of controller parameters for DC-link and battery-current regulation	Simulation + HIL	Overshoot reduced from 27.5% to 3.75%; settling time reduced from 125 ms to 20 ms	No full experimental prototype; HIL includes communication-delay effects
[20]	2025	Robust PI	SSA-based tuning	Optimization-assisted PI tuning for improved DC-link control, MPPT support, and grid-current quality	Simulation only	THD reduced to 2.50%; settling time improved to 0.015 s	Simulation only; weighting coefficients not fully disclosed
[21]	2025	Hierarchical PI	ABC-based coordinated tuning	System-wide PI retuning for bidirectional off-board fast-charger control and power-quality improvement	Simulation only	DC-link settling time reduced from 920 ms to 238 ms; THD < 1.2%	Simulation only; no HIL or experimental validation



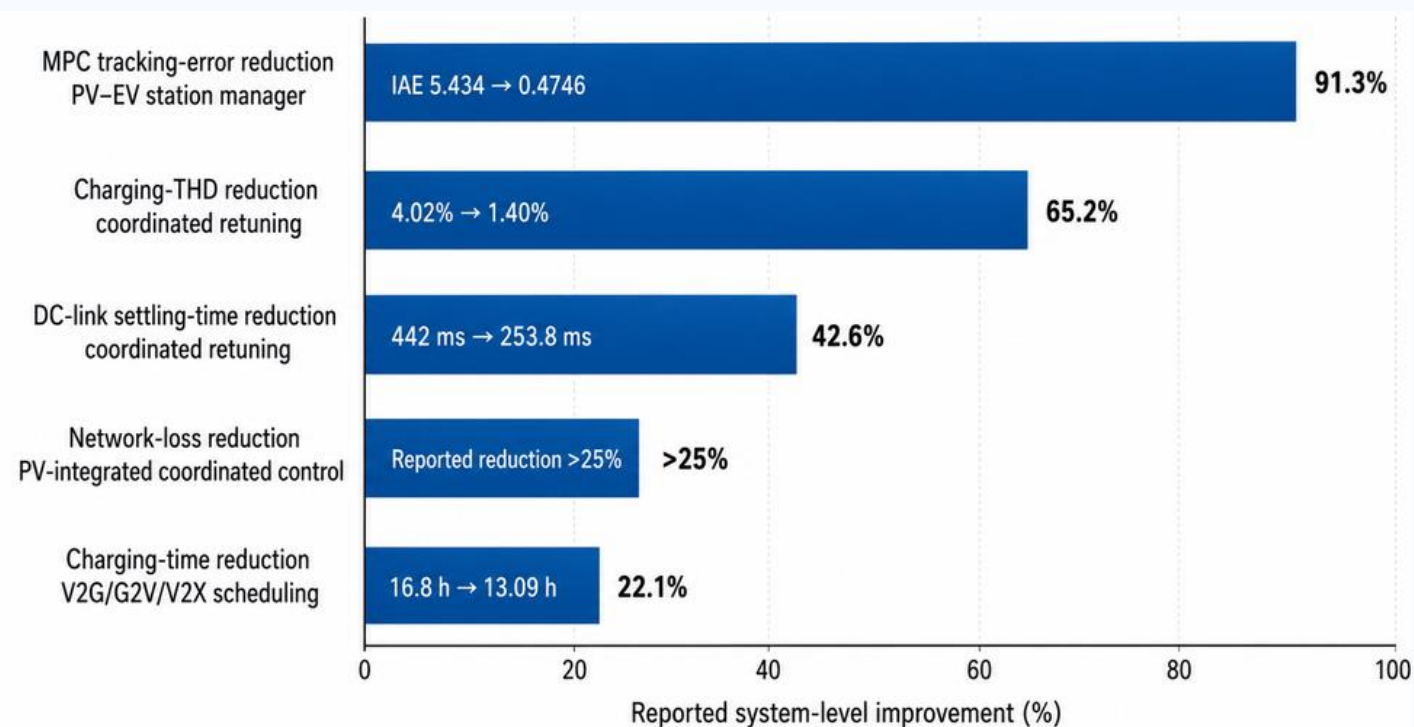
Representative gains: THD 29.301% → 2.008%; DC-link settling 920 → 238 ms.

Layer III: PV Integration Enables Coordination

PV-EV integration

TABLE III
COMPRESSED COMPARISON OF INTEGRATED PV-EV CHARGING AND ENERGY-MANAGEMENT STUDIES

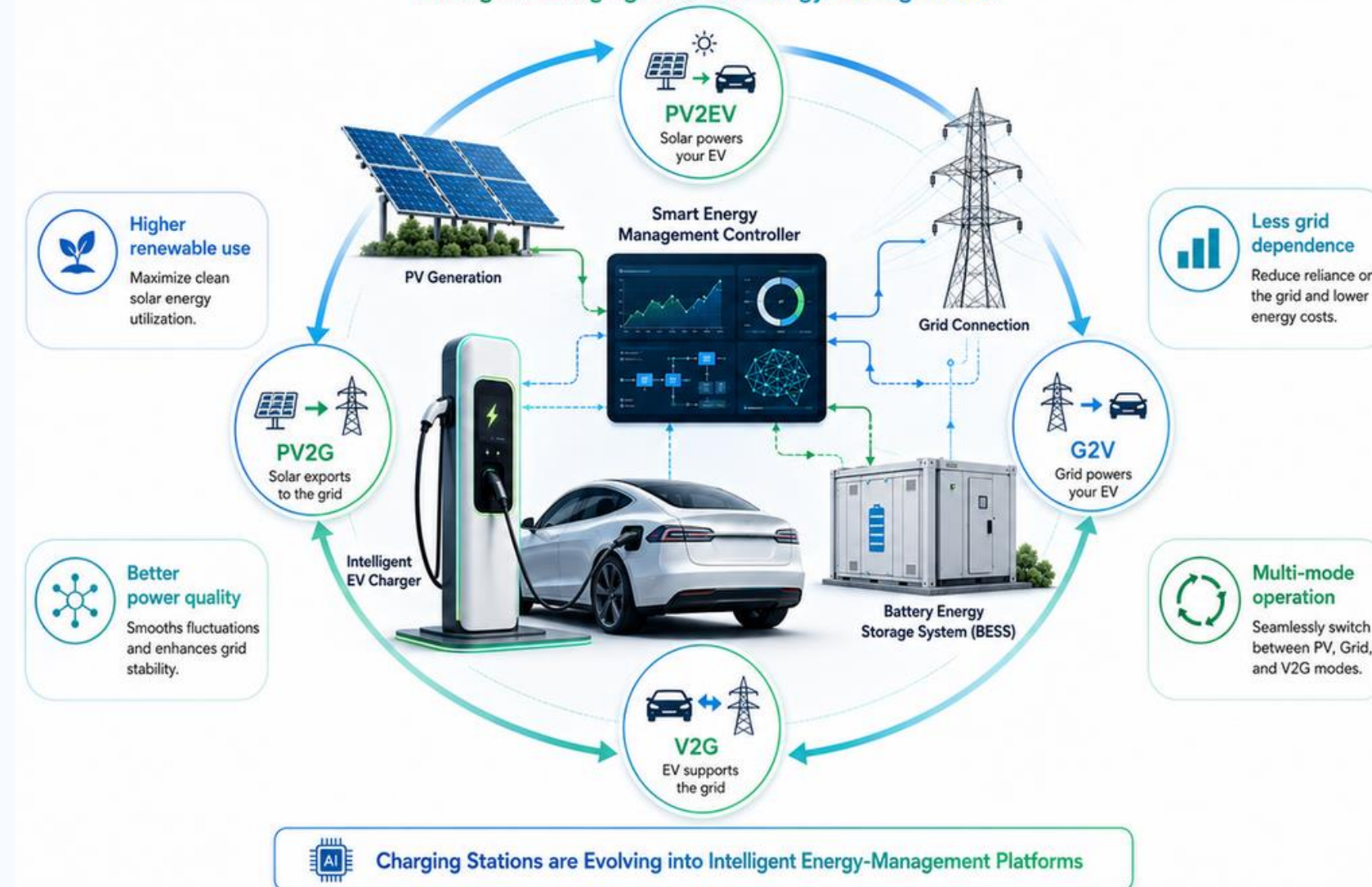
Ref.	Year	System Architecture	Management / Control Method	Principal Integration Contribution	Validation Level	Key Reported Improvement	Principal Limitation
[22]	2025	PV-integrated dual-station EV charging infrastructure	Dual-SOGI coordinated control with PI regulation	Coordinated PV-EV-grid operation for harmonic mitigation, reactive power compensation, and improved grid power quality	Simulation + experimental prototype	Grid current THD $\approx$ 1–1.1%; PF $>$ 0.98; network losses reduced $>$ 25%	No formal optimization framework; controller tuning not explicitly reported
[23]	2025	Grid-connected PV-powered EVCS with battery storage	Hybrid SWO-MHFAN energy-management framework	Hybrid prediction-and-optimization-based energy management for PV-powered EV charging stations	Simulation only	THD 1.04%; PF 0.986; operating cost 0.124 M\$/year	No experimental validation; constraints and statistical analysis not fully reported
[24]	2025	PV-grid-EV charging system with 5-level bidirectional multilevel converter	PI + P&O MPPT + SOGI-PLL	Renewable-integrated bidirectional charging with multi-mode operation and improved grid power quality	Simulation + HIL	Efficiency 97.1% (G2V), 96.7% (V2G); THD 1.2% / 0.9%	No optimization-based tuning; limited to single-phase low-power configuration
[25]	2024	Grid-connected PV-EV charging station with station manager and common DC link	Station manager + MPC	Real-time dynamic power management through adaptive on-grid/off-grid mode selection	Semi-experimental (PIL)	MPC IAE 0.4746 vs. 5.434 (PID); improved charging continuity and PV utilization	No V2G capability; implementation cost and complexity remain high
[26]	2026	Grid-integrated PV-based bidirectional off-board fast charger	Fuzzy VSS-P&O + cascaded PI + ABC retuning	System-level coordination of MPPT, grid-side control, and battery-side regulation in a PV-assisted fast charger	Simulation only	Charging THD improved from 4.02% to 1.40%; DC-link settling time 442 $\rightarrow$ 254 ms	Simulation only; optimization performed offline
[27]	2025	Conceptual V2G/G2V/V2X energy-exchange system	ABCO-based scheduling / optimization	Systems-level optimization of bidirectional and multi-entity energy exchange	Simulation only	Charging time 13.09 h vs. 16.8 h baseline; reactive power improved to -200 Var	No converter-level modeling; conceptual formulation only
[28]	2024	PV-assisted grid-connected multi-output EV charger	PWM-based mode switching	Multi-output charging architecture supporting PV2V, G2V, and V2G functions	Simulation + experimental prototype	Experimental efficiency 90.18%; total output 500 W	No closed-loop supervisory control; THD not reported
[29]	2024	Autonomous PV-powered EV charging station with storage battery	BELBIC-based intelligent control	Intelligent energy-management-oriented control for DC-bus regulation and charging-station performance enhancement	Simulation only	Overshoot reduced to $\leq$ 6%; settling time reduced to $\leq$ 0.08 s	No experimental validation; heuristic tuning only



System-Level Improvements through PV-EV Integration and Energy Management

Findings from Layer III: PV-Integrated Charging and Energy Management

Intelligent Charging. Smarter Energy. Stronger Grid.



Key insight: MPPT couples PV, DC-link, battery, and grid control.

Research Gaps and Evidence-Based Priorities

From isolated gains to deployment-ready co-design

Unresolved Gap → Research Priority

- 1 Siloed architecture/control/EMS → hierarchical co-design.
- 2 Battery ageing/electrothermal effects → lifecycle-aware objectives.
- 3 Simulation/small-scale evidence → HIL, RTDS, and full prototypes.
- 4 Idealized scenarios/uneven benchmarking → reproducible, uncertainty-aware evaluation.
- 5 Weak scaling/grid-code evidence → multi-charger, grid-responsive validation.



Conclusion

Three conclusions from the cross-layer synthesis



1

Architecture advances efficiency, gain, ripple control, voltage range, and bidirectionality - but no single topology dominates.

2

Control and PV integration shift the problem toward coupled system operation:

- Dynamic regulation
- Power-quality control
- PV/battery coordination
- Grid-responsive operation
- Supervisory energy management

3

Deployment readiness requires unified, battery-aware, reproducible, and realistically validated cross-layer co-design.

Next-generation EV charging must be scalable, adaptive, renewable-coordinated, and grid-responsive.



THANK YOU